

HALE COUNTY AP, TX, USA

WMO: 722174

Lat: 34.167N

Lon: 101.717W

Elev: 3374

StdP: 12.99

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 23097

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.6	18.2	-5.8	4.6	43.5	0.6	6.4	37.9	32.6	56.3	28.6	55.5	7.2	10	0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.8	97.2	64.4	95.2	65.9	92.8	66.4	73.4	84.2	71.5	84.2	70.3	83.0	12.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.5	127.3	77.5	68.4	118.4	75.3	67.0	112.6	74.6	39.5	83.5	37.6	83.9	36.5	81.9	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.1	24.7	20.9	DB	7.3	101.6	4.5	3.5	4.0	104.1	1.4	106.2	-1.2	108.1	-4.4	110.6
			WB	5.8	75.1	4.0	3.4	2.9	77.5	0.6	79.5	-1.6	81.4	-4.5	83.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.2	39.5	43.1	51.5	58.5	66.8	76.5	77.7	77.0	70.2	59.2	48.8	40.3	(6)
	DBStd	15.95	9.20	10.13	9.17	8.49	8.39	5.18	4.28	4.46	5.84	8.48	8.84	9.14	(7)
	HDD50	1161	340	232	92	24	4	0	0	0	0	24	127	318	(8)
	HDD65	3617	789	613	424	223	80	1	0	0	20	214	488	765	(9)
	CDD50	4509	16	39	139	280	524	794	859	838	605	308	89	18	(10)
	CDD65	1490	0	0	5	29	136	345	394	373	175	32	1	0	(11)
	CDH74	17441	8	46	214	669	1865	3962	4330	3954	1792	510	85	6	(12)
	CDH80	8107	1	6	40	211	845	2025	2156	1974	708	128	13	0	(13)
(14) Wind	WSAvg	10.0	10.0	10.8	11.5	12.5	11.2	11.0	8.4	7.8	8.2	9.2	9.7	9.9	(14)
(15) Precipitation	PrecAvg	20.0	0.6	0.6	0.9	1.3	2.9	3.3	2.5	2.3	2.2	1.7	0.9	0.7	(15)
	PrecMax	34.1	2.6	2.2	3.0	5.7	8.5	10.0	11.7	8.0	5.4	6.5	5.6	2.6	(16)
	PrecMin	8.9	0.0	0.0	0.0	0.0	0.2	0.3	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	5.6	0.6	0.6	0.8	1.2	1.9	2.2	2.2	1.7	1.6	1.8	1.0	0.6	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.4	79.7	84.0	90.4	97.1	100.4	99.7	98.8	95.1	89.7	81.8	73.4	(19)
		MCWB	47.2	51.0	52.8	55.6	59.4	62.2	65.2	65.8	65.3	60.2	53.2	48.6	(20)
	2%	DB	69.7	74.3	79.7	85.8	94.6	97.3	96.6	95.5	90.9	83.1	76.2	68.0	(21)
		MCWB	47.7	49.5	53.2	54.1	59.1	63.4	66.3	67.7	65.1	60.3	52.2	47.8	(22)
	5%	DB	62.8	69.2	75.3	81.6	90.0	95.0	94.6	94.5	89.5	79.7	71.7	61.5	(23)
		MCWB	44.2	48.0	51.7	53.9	59.0	66.0	67.8	68.0	65.2	60.1	50.9	44.6	(24)
	10%	DB	57.1	62.4	70.4	77.4	84.0	90.8	90.6	90.4	83.6	75.8	65.8	56.6	(25)
		MCWB	42.4	44.3	50.7	54.4	59.6	65.9	68.4	68.2	64.4	57.4	49.3	43.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.5	54.5	60.8	64.2	69.2	74.9	79.5	73.8	71.4	68.8	59.0	53.8	(27)
		MCDB	63.3	68.7	72.1	74.8	79.8	84.1	87.6	88.2	83.2	77.7	68.7	60.0	(28)
	2%	WB	48.7	51.6	57.7	61.4	67.1	72.3	75.1	72.0	69.7	66.1	56.2	50.1	(29)
		MCDB	64.4	69.4	68.2	72.9	78.1	84.5	83.3	86.3	80.3	74.9	65.8	61.4	(30)
	5%	WB	45.7	49.0	55.3	59.3	65.5	70.7	72.0	70.8	68.6	64.0	53.8	47.3	(31)
		MCDB	60.9	66.3	67.6	69.6	76.9	83.7	82.6	84.9	79.2	72.5	65.3	60.7	(32)
	10%	WB	42.9	45.9	52.8	56.9	63.6	69.3	70.3	69.5	67.4	61.4	51.2	44.0	(33)
		MCDB	56.8	61.1	66.1	69.5	76.0	82.2	82.0	83.2	78.1	71.2	64.5	57.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	27.5	28.0	29.2	29.0	26.4	25.7	23.8	24.5	23.9	26.5	27.7	25.2	(35)
		MCDBR	36.7	37.9	36.6	36.2	34.0	30.6	28.3	28.3	30.7	32.1	36.0	33.9	(36)
	5% WB	MCWBR	19.7	19.1	17.2	15.2	12.9	9.9	8.7	9.1	10.7	13.9	16.8	18.2	(37)
		MCWBR	34.1	35.5	29.3	27.7	26.1	25.1	22.2	24.4	22.7	24.2	29.3	30.5	(38)
(40) Clear-Sky Solar Irradiance	taub		0.258	0.268	0.295	0.327	0.340	0.380	0.384	0.373	0.343	0.311	0.282	0.264	(40)
		taud	2.587	2.567	2.499	2.405	2.433	2.364	2.400	2.393	2.454	2.523	2.536	2.565	(41)
	Ebn at Noon		309	315	312	304	299	286	284	286	291	295	296	300	(42)
		Edh at Noon	24	28	32	37	36	39	37	37	33	29	25	23	(43)
(44) All-Sky Solar Radiation	RadAvg		1042	1289	1688	2064	2241	2384	2293	2087	1747	1373	1088	898	(44)
	RadStd		95	119	119	141	181	141	102	96	112	153	109	67	(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (13 neighbors)	+0.52	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+175	N/A	(47)

Nomenclature: See separate page